DATASCI 406 FA 26 — Module 1 Mock Exam

Name and Uniqname:

This practice exam asks about the Day 5 Benford's Law exercises. The real exam asks the same kinds of question about your project, so work this one with your worked exercises in front of you — every question points at code you have already run, though a few ask you to think past what you ran. Exercise numbers refer to the session exercise document.

No question asks you to copy a number out of your output. They ask why you did it that way, what would happen if you changed it, and where in the exercises a given technique appears.

50 points. Budget about 62 minutes.

Part A — Monte Carlo integration (10 points)

A1 (2 points)

In the exercise that draws F uniformly on $(0, 1)$, you simulated 10,000 values of X and reported the proportion carrying each leading digit. The proportion for $d = 9$ came out about 0.049, against $w_9 = 0.046$. You want a confidence interval for that 0.049. **Circle one.**

1. A binomial interval on the count of nines among the 10,000 draws.
2. A t interval across the nine proportions in the table.
3. None is needed — w_9 is exact, so the 0.049 carries no error.
4. A binomial interval on the count of nines out of 9.

Solution

Answer: 1. The entry is the mean of $\mathbf{1}(D_i = 9)$ over $n = 10{,}000$ independent draws — a binomial proportion out of n.

2 averages nine dependent numbers that sum to 1. 3 confuses the exact target with the simulation of it — w_9 is exact, the 0.049 is not. 4 uses the number of digits as the number of trials.

A2 (2 points)

Your table of digit proportions came from 10,000 draws, and its entries are good to about two decimal places. You want a third. Roughly how many draws? **Circle one.**

1. 20,000
2. 100,000
3. 1,000,000
4. No number of draws will do it; the third decimal is beyond Monte Carlo.

Solution

Answer: 3. The error falls like $n^{-1/2}$, so a tenfold gain in precision costs a hundredfold in draws. At $n = 10^4$ the standard error on a proportion near 0.3 is 0.0046; at $n = 10^6$ it is 0.00046.

This rate is the one thing about Monte Carlo we cannot negotiate — variance reduction buys a better constant, never a better exponent. 4 confuses a slow rate with a floor.

A3 (6 points)

You built two samplers for $X = 10^{M+F}$. The first picks one of the nine intervals I_d uniformly and then F uniformly within it; the second takes $F \sim U(0, 1)$ directly. Both draw $M \sim \text{Poisson}(2)$.

(i) What changes in the table of digit proportions if M is drawn from Poisson(50) instead, or dropped entirely so that $X = 10^F$? Why?

(ii) The first sampler gives uniform digits and the second gives Benford digits. Name the single probabilistic difference responsible, stated in terms of the intervals I_d.

(iii) $P(D = d)$ has the closed form w_d. So what is the second simulation *for*? Say what agreement between the table and w_d establishes, and what a disagreement would mean.

Solution

(i) Nothing. M is an integer, so 10^M only moves the decimal point; the leading digit depends on F alone.

(ii) Which interval gets chosen. The first gives every I_d probability $1/9$; the second gives it w_d, its width.

(iii) It checks the code and the claim $P(F \in I_d) = w_d$ against an answer already known. A gap beyond Monte Carlo error means one of the two is wrong, but not which.

Rubric. 2 pts each. (ii) needs the probability assigned to I_d, not the widths — both samplers use the same unequal widths.

Part B — variance reduction (10 points)

B1 (2 points)

To estimate $P(D = 1) = \log_{10} 2$ you draw $F \sim U(0, 1)$ and average $\mathbf{1}(F < \log_{10} 2)$. A classmate proposes antithetic pairs $(F, 1 - F)$, averaging within each pair. **Circle one.**

1. Invalid — the two draws in a pair are not independent.
2. Valid, and it cuts the variance by about 43% at a matched budget.
3. Valid, but it can never help when the quantity averaged is an indicator.
4. Valid, and it cuts the variance by about the same amount for every digit d.

Solution

Answer: 2. The indicator is monotone in F, so the pair is negatively correlated: F and $1 - F$ cannot both fall below $\log_{10} 2$, giving $\rho = -p/(1 - p) = -0.43$. The reduction at matched budget is $-\rho$.

1 mistakes the mechanism for a flaw — pairs are still iid. 4 fails at $d = 9$, where $p = 0.046$ and the reduction is 5%.

B2 (2 points)

You try the antithetic pairing $(F, 1 - F)$ for every digit in turn. For one of them it makes the variance *worse*. Which, and why? **Circle one.**

1. $d = 1$, because I_1 is the widest interval.
2. $d = 3$, because I_3 is the interval containing 0.5.
3. $d = 9$, because I_9 is the narrowest interval.
4. None; the pairing can never do worse than independent draws.

Solution

Answer: 2. $I_3 = [0.477, 0.602)$ is the one interval straddling 0.5, so F and $1 - F$ can both land in it. The two indicators are then *positively* correlated — $\rho = +0.276$, making the variance 28% worse than independent draws at the same budget.

Every other digit's interval is disjoint from its own reflection, so both indicators cannot fire at once and $\rho = -p/(1-p) < 0$: a large help at $d = 1$, a small one elsewhere. 4 is the tempting general claim and it is false; antithetic sampling is only guaranteed not to hurt when h is monotone.

B3 (6 points)

(i) Exercise 6 used $B = 4000$ and returned $p = 1/4001 \approx 0.00025$, with no null draw anywhere near the observed distance. A classmate wants to report $p = 10^{-5}$ instead. What would they have to change, and by how much? Would more FEC contributions let them report a smaller number?

(ii) The power exercise reported 0.379 at $n = 50$, from 1,000 replicates. Give the Monte Carlo standard error of that number, and say how many replicates would halve it.

(iii) That exercise spends its budget twice: one batch of replicates to set the critical value, then a fresh batch from the alternative to estimate the power. If you could afford 1,000 replicates *in total*, how would you split them? Say what goes wrong at each extreme.

Solution

(i) Raise B: $1/(B+1) \le 10^{-5}$ needs $B \ge 99{,}999$. More contributions would not let them report a smaller number — the floor is set by B alone. They would be stronger *evidence*, but n is fixed at 8,396 either way.

(ii) $\sqrt{0.379 \times 0.621/1000} = 0.015$. Halving it takes four times the replicates, 4,000.

(iii) Roughly even. All on the cutoff leaves no power estimate; all on the alternative leaves a cutoff from too few null draws. An error in the cutoff shifts every one of the power comparisons the same way, so it acts as bias rather than noise.

Rubric. 2 pts each. (iii) is graded on the bias-versus-noise asymmetry, not on the split — a student who argues from it for weighting the null batch has given the better answer and earns full credit.

Part C — Monte Carlo for estimation (10 points)

C1 (2 points)

The products exercise simulated 100,000 values of $X = \prod_{i=1}^{k} V_i$ for $k \in \{1, 5, 25, 100\}$. You want the leading digits in the table to sit closer to w_d. **Circle one.**

1. Raise the number of simulated values.
2. Raise k.
3. Either one; both shrink the gap between the table and w_d.
4. Neither; the gap is fixed by the choice of Exp(10) and cannot be changed.

Solution

Answer: 2. k is a feature of the thing simulated — the true digit distribution moves toward Benford as k grows. The replicate count is the simulation budget, fixing how precisely the table estimates whatever is true at that k.

At $k = 1$ the truth is 0.951, and 10^9 replicates would report 0.951 precisely.

C2 (2 points)

Exercise 6 simulates the null distribution with 4,000 replicates. What is *one replicate*? **Circle one.**

1. One simulated leading digit.
2. One of the nine simulated digit proportions.
3. A whole simulated dataset of 8,396 digits, reduced to a single distance.

4. The observed FEC data, resampled.

Solution

Answer: 3. The statistic is a function of an entire dataset, so producing one draw from its sampling distribution means generating a dataset the size of the real one and summarizing it. The histogram has 4,000 points and rests on 4000×8396 simulated digits.

1 and 2 are pieces of a replicate, not a replicate. 4 describes a bootstrap, which resamples the observed data; here we sample from the null distribution instead, which is fully specified and needs no data.

C3 (6 points)

Treat the vector of nine observed digit proportions, computed from a sample of size n, as an estimator of the true digit distribution.

(i) The $k = 1$ row puts digit 1 at about 0.95 against $w_1 = 0.301$. Is that bias in the estimator? Answer in terms of what the estimator is estimating.

(ii) In the $k = 100$ row the entries differ from w_d in the third decimal place. What would you compute to decide whether those differences are real, and roughly what value does it take for the $d = 1$ entry?

(iii) Summarize a row by the distance statistic T — the Euclidean distance between the nine observed proportions and w_d. If the truth really is Benford, is $\mathrm{E}(T) = 0$? Say why, and say what happens to the null distribution of T as n grows.

Solution

(i) No. The estimator is unbiased for the truth it is handed, and at $k = 1$ that truth is not Benford: $Y \sim \mathrm{Exp}(10)$ has mean 0.1, so F falls below 0.301 almost always. The gap is a fact about the distribution.

(ii) $\sqrt{\hat{p}(1-\hat{p})/n} = \sqrt{0.301 \times 0.699/10^5} = 0.0014$, or a binomial interval on that entry. A third-decimal gap is about one standard error.

(iii) No. $T \geq 0$ and is zero only if every $\hat{p}_d$ lands exactly on p_d, so $\mathrm{E}(T) > 0$ under the null — the Exercise 6 null has mean 0.0096. Each $\hat{p}_d$ has SD of order $n^{-1/2}$, so the null distribution slides toward 0 and tightens as n grows.

Rubric. 2 pts each. (iii) needs both $\mathrm{E}(T) > 0$ and the $n^{-1/2}$ rate.

Part D — Monte Carlo hypothesis tests (10 points)

D1 (2 points)

The power exercise takes the critical value to be the 95% quantile of a batch of simulated null statistics, then draws a **fresh** batch from the alternative to estimate the power. Suppose you also wanted to check the size, and did it by asking what fraction of *those same null statistics* exceed the critical value. What do you get? **Circle one.**

1. About 0.05, confirming the test is correctly sized.
2. Exactly 0.05 up to rounding, by construction — the check establishes nothing.
3. Something well above 0.05, because reusing draws inflates the size.
4. Something well below 0.05, because a sample quantile is conservative.

Solution

Answer: 2. The cutoff is the 95% quantile of those very numbers, so 5% of them exceed it whatever was drawn. A size check needs fresh null draws, just as the power estimate uses fresh alternative draws.

1 gets the number right and the reason wrong, which is the whole question.

D2 (2 points)

Exercise 7 takes the cutoff to be the 95% quantile of the simulated null statistics, at each n separately. What does that choice fix? **Circle one.**

1. The power, at 0.95.
2. The size, at 0.05, whatever n is.
3. The p-value for the observed data.
4. Nothing, until we see how many alternative draws exceed it.

Solution

Answer: 2. Taking the 95% point of the null puts 5% of null datasets above the cutoff by construction — that is the size, and it holds at every n because the quantile is recomputed each time. The cutoff itself changes a great deal (0.181 at $n = 50$, 0.015 at $n = 8396$); the size it delivers does not.

Fixing the size first is what makes the three rows comparable: power is only meaningful against a common α. 1 inverts size and power. 4 is the error D1 turns on — the alternative draws measure power, not size.

D3 (6 points)

The test in Exercises 6–7 rejects for large T only.

(i) Why is a lower tail not part of the rejection region? Say what it would cost and what it would buy, given the alternative Exercise 7 considers.

(ii) Describe a situation in which a *small* T would be the suspicious outcome. What is the alternative hypothesis there?

(iii) A collaborator runs your test on 50 contributions, does not reject, and writes "these digits are consistent with Benford's Law, so the filings are clean." Using the Exercise 7 table, say what is wrong and what you can say instead.

Solution

(i) It would spend α on the lower tail and buy nothing. Any alternative digit distribution adds a positive systematic term to T, so none of them makes T unusually small.

(ii) Data fabricated to pass — digit counts set proportional to w_d, or a table rounded to match, fit closer than sampling noise allows. The alternative is that the digits are not iid draws from anything.

(iii) Not rejecting is not accepting. Power against the halfway-to-uniform alternative at $n = 50$ is 0.379, so it slips past three times in five. The same test at $n = 8396$ has power 1.000, where a non-rejection would mean something.

Rubric. 2 pts each. (iii) needs a number from the table. (ii) earns 1 for "fits too well", the second for naming what produces it.

Part E — the whole module (10 points)

(a) Three sample sizes (5 points)

Day 5 carries three counts that all look like a sample size. The letters standing for them have not been consistent across the module, so here they are by description:

- **the data** — 8,396 contributions in Exercise 6

- **the replicates** — 4,000 null draws in Exercise 6; in Exercise 7, 1,000 null and 1,000 alternative draws for each row of the table
- **the factors** — the $1, 5, 25, 100$ values multiplied together in Exercise 4

(i) One sentence each: what does each of the three control?

(ii) The Exercise 6 p-value is reported as $< 1/4001$. Which of the three would you raise to report a *smaller number*, and which would you raise to make the *conclusion more convincing*? These need not be the same.

(iii) Suppose the filings had held 200 contributions rather than 8,396, and the digit proportions came out about as they did. What happens to the p-value, and why? Name one number in Exercise 6 that does *not* move.

Solution

(i) The data is evidence about the world and not ours to choose; it fixes how small a departure the test can detect. The replicates are simulation budget, fixing how precisely we know the null distribution and the p-value, and costing only computer time. The factors are a property of the process in Exercise 4 — multiplying more of them moves the truth itself toward Benford.

(ii) Replicates for a smaller number, since the p-value cannot fall below $1/(B+1)$. Nothing, for a better conclusion: no null draw exceeded 0.023 against an observed 0.135, so refining 0.00025 adds nothing. Only more contributions would sharpen the evidence, and 8,396 is what exists.

(iii) The p-value rises and leaves the floor. The observed distance is a function of the nine proportions alone, so it stays near 0.135 — but the null distribution widens with less data: Exercise 7 puts its 95% point at 0.092 for $n = 200$ against 0.015 at $n = 8396$, so 0.135 is no longer far beyond every null draw. Unmoved: the observed distance itself, w_d, $B = 4000$ and the floor $1/4001$.

Rubric. 2 pts (i), 2 pts (ii), 1 pt (iii). (ii) needs the distinction — replicates buy resolution, data buys evidence. In (iii), "the observed distance changes" earns nothing: the question holds the proportions fixed.

(b) Find an example (5 points)

Point to one place in the Day 5 exercises that does each of the following, and say in a sentence why it fits.

1. Estimating a probability or expectation with no closed-form answer.

2. Approximating the sampling distribution of a statistic with no tractable form.

3. Measuring an operating characteristic of a testing procedure.

4. Investigating a claim you have no formal proof of.

5. Checking a simulated answer against something you already know exactly.

Solution

1. The Exercise 6 p-value, $P(m \geq m_{\text{obs}})$ under the null.

2. The 4,000 null distances in Exercise 6; Exercise 7 reads its critical values off the same object.

3. Exercise 7's power against the flatter alternative.

4. Exercise 4: that the digits approach w_d as k grows. The CLT argument is a heuristic about a limit; the simulation asks when it arrives.

5. Exercise 3 against w_d, and Exercise 2 against the $1/9$ its construction guarantees.

Rubric. 1 pt each, generously. One exercise may serve more than one item.